

PDS Joint: A Parametric Double-Spiral Joint Tailored for Dexterous Hands

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Abstract—Compliant joints can embed safety and adaptability into dexterous hands, but achieving large-stroke anthropomorphic motion while maintaining joint-specific, direction-dependent stiffness and reliable proprioception remains challenging. This paper presents the *PDS joint*, a parametric double-spiral compliant joint that enables systematic shaping of directional stiffness across multiple deformation modes, including flexion/extension, abduction/adduction, and pronation/supination. We instantiate the joint using Archimedean and logarithmic spiral templates for different hand joints and introduce an asymmetry ratio to tailor stiffness distributions for both grasp stability and hyperextension resistance. To make the joint practically usable under large deformation, we co-design embedded inductive proprioception and propose a learning-based calibration pipeline that maps raw inductive signals to joint states using ArUco-marker tracking. Experiments characterize the stiffness landscapes across geometric parameters and demonstrate a non-monotonic dependence of lateral support on asymmetry, indicating the importance of principled parameter tuning. For joint-state estimation in the most challenging abduction/adduction motion, an MLP-based mapping reduces error compared with conventional curve fitting by 59.2%. Finally, we integrate the proposed joints into an open-source dexterous hand as a demonstration platform and achieved full success rate in adaptive grasping tests and safe, contact-rich human-involved interactions.

Keywords: compliant joint, dexterous hand, spiral mechanism, proprioception, inductive sensing, calibration

I. INTRODUCTION

From simple grippers to dexterous hands, robotic end-effectors are rapidly evolving along with embodied AI. However, safe and robust deployment in the real world remains a limiting factor, especially in human-robot interaction (HRI), high-speed motion, and cluttered environments, where uncertain impacts and disturbances are inevitable.

Many dexterous hands adopt predominantly stiff joints to achieve high position accuracy. However, these designs often come with increased distal inertia and tend to amplify the consequence of unexpected collisions or joint over-travel, where hard mechanical stops can cause large impulsive forces on both the robot and its environment. Moreover, stiff joints commonly require complicated assembly and often rely on accurate sensing and robust algorithms to maintain performance under contact-rich circumstances, increasing system complexity and software burden. These challenges motivate joint designs that can inherently tolerate uncertainty and impacts, while remaining accessible to scalable manufacturing and maintenance.

Compliant mechanisms are a compelling alternative because they embed mechanical intelligence into the joint, allowing passive adaptation and absorbing unexpected impacts.

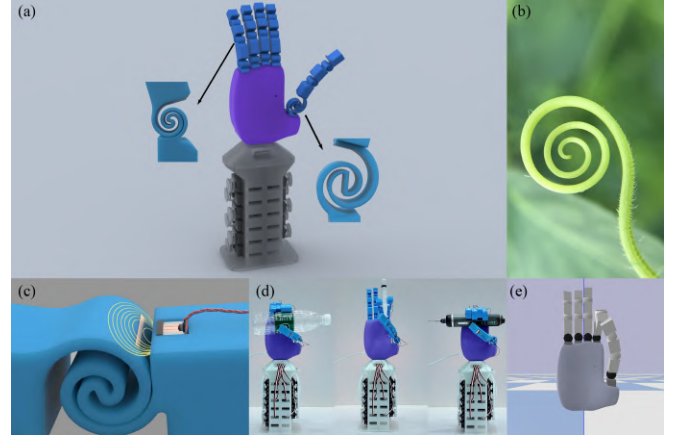


Fig. 1. (a) An overview of the PDS-joint-based dexterous hand platform. (b) Demonstration of the spiral morphology in fern fronds. (c) Embedded inductive sensing module. (d) The compliant hand performs grasp and manipulation tasks. (e) A synchronized digital twin enabled by proprioception.

Previous works have explored directly using elastomers in underactuated hand designs [1]. However, the viscoelasticity of elastomers causes their effective stiffness and damping to drift with time and temperature, degrading repeatability across tasks [2]. Flexure-based joints can reduce backlash and maintenance effort by replacing rigid components with monolithic elastic parts, yet they may lose support stiffness in load-bearing directions at large deflections, reducing grasp stability under lateral disturbances [3]. Rolling-contact joints provide another route to large-range motion and can enable directional stiffness shaping [4], but typically require additional parts and assembly, increasing design complexity [4]. Most importantly, many compliant joint designs explored for robotic hands are dominated by a single bending mode, which makes it nontrivial to match the varying degrees of freedom (DoF) of human finger joints, limiting their applicability beyond simple grippers.

Nature offers elegant solutions. In many fern fronds, tightly coiled fronds provide compact reversible deformation and effectively protect the leaf apical meristem during gradual unfurling [5], [6]. Inspired by such spiral morphologies, spiral-based compliant elements provide a compact geometric template for large-stroke deformation, where mechanical responses can be tuned by a simple set of geometric parameters [7], [8]. However, existing double-spiral studies primarily introduce the structure as a general-purpose building block for engineering design. For application in dexterous hand joints, what remains missing is a detailed investigation

into how the parameter changes shape the mechanical behaviors and the directional stiffness distribution that could be tailored for different joints in dexterous hands.

Notably, this gap is not purely mechanical. Complex deformation in compliant finger joints is difficult to be accurately modeled, limiting their applications in dexterous hands when complex tasks are involved. Recent studies suggest that despite the adaptability, compliant hands still demand reliable proprioception in complex tasks [9]. Hence, a compact proprioception system integrated with the compliant joint is necessary. Surveys have shown that resistive, optical, capacitive, and magnetic sensors each have distinct limitations in compliant joint integration [10]. Based on these observations, we integrated an inductive sensing system, which is robust against strong EMC disturbances and stray magnetic fields, with the spiral joint, and proposed a practical learning-based calibration pipeline using ArUco markers. [11]–[13].

Therefore, we present the *PDS joint*, a **parameterized double-spiral joint** that targets the joint-specific stiffness requirements in dexterous hands. To make such compliant joints practical in complex tasks, we further co-design the joint with **embedded inductive proprioception** and introduce a **learning-based calibration pipeline** that maps raw inductive signals to joint states using ArUco markers. Finally, we integrate the proposed joints into an open-source dexterous hand as a *demonstration platform* and validate both the mechanical characteristics and the interaction robustness (Fig. 1).

The key contributions are:

- **A parameterized double-spiral compliant joint for dexterous hands** that supports large reversible deformation and multi-mode motion, while enabling joint-specific *directional* stiffness shaping (flexion/extension, abduction/adduction, pronation/supination) through a simple geometric parameter set.
- **A systematic parameterization study** that characterizes how geometric parameters influence directional stiffness distribution, providing practical design guidelines for tailoring different finger joints within a dexterous hand.
- **An integrated inductive proprioception system for the double-spiral joints** that is compact, contactless, and robust to EMC disturbances and stray magnetic fields, suitable for real-world contact-rich manipulation [11]–[13].
- **A practical learning-based calibration pipeline** that maps inductive signals to joint states under large deformation, allowing accessible and reliable state estimation for pinch grasping and contact-rich tasks. [9].

II. RELATED WORK

A. Compliant joints for anthropomorphic hands

The OpenHand family [1] adopts a hybrid deposition manufacturing (HDM) process, which improves the robustness and impact resistance of the compliant hinge, but

comes with viscoelastic drift and hysteresis, which hinder repeatable joint behavior and state estimation across time and temperature [2].

For large-stroke applications, Naves *et al.* show that the support stiffness in load-bearing directions can drop markedly with deflection, exposing a severe conflict that must be addressed by topology design [3]. Garcia *et al.* further compare cross-hinge topology families for anthropomorphic finger joints and quantify support stiffness over the full range of motion, highlighting off-axis stability as a primary bottleneck [14].

Rolling-contact joints pursue large-range anthropomorphic kinematics with low friction. ARC Joint introduces tongue-and-groove contact with collateral-ligament-like elastic elements to obtain sufficient torsional stiffness for pinch stability [4]. Such performance typically relies on additional elastic components and tight contact constraints to maintain stable rolling motion, which complicates the design and assembly, and challenges compact integration in multi-fingered hands [4].

Overall, prior compliant joints predominantly focus on a single-DoF design or lack a generalizable geometric paradigm that can be applied to different joints across the hand according to the anthropomorphic stiffness distribution, motivating a systematic parameterized design.

B. Spiral-inspired compliant elements

Spiral springs have long been used as compact energy-storing elements with large deformation. Kim *et al.* proposed a dual-spiral-spring actuation system (DSSAS) for wearable robots, leveraging the spiral spring's high compliance and large deformation. [15]. Radaelli and Herder conducted an exploratory study on large-displacement spiral springs and demonstrated how varying cross-section properties, material orthotropy, and prestress changes the torsional stiffness, providing design insights for stiffness tailoring [16].

Building on spiral templates, the double-spiral geometry couples two spirals into a compliant element with multiple degrees of freedom and tunable behaviors [7], [8]. While prior studies establish a versatile library of large-stroke compliant elements, they typically focus on general-purpose functionalities or single target loading directions. For dexterous hand joints, what remains missing is a systematic parameter-to-mechanics mapping that links geometric parameters to mechanical behaviors, enabling joint-specific stiffness configuration (flexion/extension, abduction/adduction, pronation/supination).

C. Proprioception and calibration for compliant joints

Recent studies suggested that precise tasks benefit from reliable state feedback rather than pure open-loop control. A line of work estimates joint behavior from measurements of actuators. For instance, a minimal-sensing flexure gripper combines displacement sensing with force estimation via direct force sensors, SEA deflection, or actuator current models [17]. However, this indirect joint state measurement

becomes unreliable with the accumulated error through actuation transmission.

Resistive strain sensors are attractive for their simplicity and ease of integration, yet hysteresis, and temperature dependence commonly lead to drift and errors in large-strain compliant mechanisms [2], [10]. Capacitive sensors can provide high sensitivity consumption, but are susceptible to parasitic capacitance, proximity effects, and electromagnetic interference, often requiring careful shielding [10], [18]. Magnetic-based sensors are compact and widely used, but in dexterous hands they can be disturbed by stray magnetic fields from motors and the environment. Mitigating this typically requires careful design or additional compensation [10], [11]. Optical approaches offer high resolution and strong immunity to electromagnetic noise, but can be sensitive to routing constraints and bending-induced signal loss in compact multi-joint hands [10]. Inductive sensing enables compact, contactless measurement using coils and conductive targets, and is widely adopted for robust non-contact position sensing with strong EMC tolerance [12], [13]. Magnet-free inductive designs further improve robustness against stray magnetic fields, which is particularly attractive for operating in electromagnetically noisy environments [11], [19]. While conventional model-based calibration can be sensitive to manufacturing errors and nonlinear deformation, data-driven calibration with external supervision is widely used in robotics. Fiducial markers such as ArUco provide reliable pose measurements for supervision even under partial occlusion [20]. These observations motivate the co-design and integration of the joint and the proprioception system, along with a robust calibration pipeline.

III. APPROACH

A. Overview

We propose the *PDS joint*, a parameterized double-spiral compliant joint intended for dexterous hands with joint-specific stiffness distribution. The joint geometry is controlled by a compact parameter set, which allows systematic shaping of directional stiffness across different joint DoFs (flexion/extension, abduction/adduction, and pronation/supination). To make the joint practically usable under distributed deformation, we co-design the joint with embedded inductive proprioception and learn a calibration mapping from raw inductive signals to joint states using vision-based supervision with ArUco markers. Finally, we integrate the proposed PDS joints into an open-source dexterous hand as a demonstration platform.

B. PDS Joint Design

Depending on degrees of freedom (DoF) and desired dynamics, we employ two spiral curves.

Archimedean spiral. For the distal interphalangeal (DIP), proximal interphalangeal (PIP), and metacarpophalangeal (MCP) joints of the four fingers, which requires compact design, we use an Archimedean spiral (Fig. 2):

$$r(\theta) = r_0 + k\theta, \quad \theta \in [0, \theta_{\max}]. \quad (1)$$

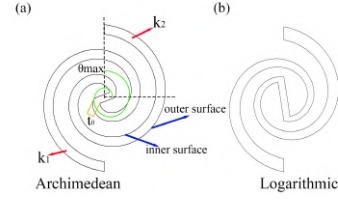


Fig. 2. (a) An Archimedean spiral and the correlating geometric parameters. (b) A logarithmic spirals.

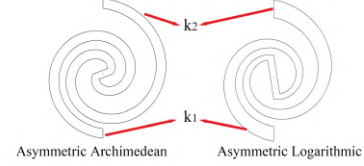


Fig. 3. Definition of the asymmetry ratio λ .

Logarithmic spiral. For the carpometacarpal (CMC) joint of the thumb, which requires larger 3D deformation for opposition, we adopt a logarithmic spiral (Fig. 2):

$$r(\theta) = r_0 e^{k\theta}, \quad \theta \in [0, \theta_{\max}]. \quad (2)$$

Here, r_0 is the initial radius, k is the polar slope, and $\theta_{\max}$ defines the angular range.

The initial thickness is defined as

$$t_0 = r_{0,\text{outer}} - r_{0,\text{inner}}. \quad (3)$$

To introduce asymmetry, different slopes are assigned to the two spiral branches, and the asymmetry ratio is defined as

$$\lambda = \frac{k_2}{k_1}, \quad (4)$$

where k_1 and k_2 are the polar slopes of opposite branches (Fig. 3). A symmetric joint corresponds to $\lambda = 1$.

For both types of joint, the geometric parameter set can be concluded as follows.

$$s = \{k_1, k_2, \theta_{\max}, r_{0,\text{inner}}, t_0, \lambda\}. \quad (5)$$

The final parameters selected for the hand platform are listed in Table I.

C. Embedded Inductive Proprioception

The sensing module includes a customized Printed Circuit Board (PCB), chip inductors, copper foil targets, LDC1614 inductance-to-digital converters [21], and an STM32F405 microcontroller. The sensor exploits eddy-current effects in an LC tank [22]. As the conductive copper foil approaches

TABLE I
SELECTED GEOMETRIC PARAMETERS FOR THE FINAL SPIRAL JOINTS.

Joint	Spiral	$\theta_{\max}$ (rad)	t_0 (mm)	k_1	λ
DIP	Archimedean	2.48π	1.4	0.75	1.2
PIP	Archimedean	2.5π	1.4	0.75	1.18
MCP	Archimedean	2.5π	1.4	0.8	1.22
CMC	Logarithmic	2π	1.5	0.18	1.12

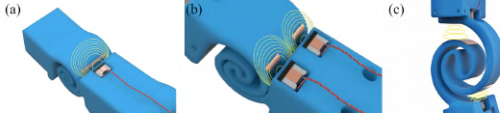


Fig. 4. Sensor placement for different joints: DIP/PIP, MCP, and CMC.



Fig. 5. Learning-based calibration using ArUco markers and synchronized sensor acquisition.

the inductor, stronger eddy currents are induced in the target and oppose the coil's magnetic field, which reduces the coil's effective inductance. Consequently, the effective inductance L varies nonlinearly but monotonically with the relative distance d . This distance-dependent inductance directly modulates the resonant frequency of the LC tank:

$$f_{\text{res}}(d) = \frac{1}{2\pi\sqrt{L(d)C}}, \quad (6)$$

where C is the parallel capacitor.

In practice, the copper foil is attached to the spiral surface and the inductor is embedded in the adjacent phalanx. As the joint bends, the relative distance between the foil and the inductor changes, producing a frequency response correlated to the joint state.

The sensor placement is tailored to the joint DoF. In particular, there is one channel for DIP/PIP flexion, two channels for MCP to decouple flexion and adduction/abduction via differential signals, and two sensor pairs on opposite sides for the thumb CMC to cover its complex 3D kinematics. The corresponding sensor placements for DIP/PIP, MCP, and thumb CMC are illustrated in Fig. 4.

D. Learning-based Calibration

To effectively calibrate sensors and map sensor signals to joint states, a calibration pipeline was developed. To start with, sensor signals and joint poses are aligned through three ROS 2 nodes:

- *Pose detection node*: uses a calibrated smartphone camera (30 FPS) via iVCam to estimate 6D poses through ArUco markers [20] pasted on the finger as shown in Fig. 5 with timestamps.
- *Signal receiving node*: reads inductive signals from serial (100 Hz) with timestamps.
- *Aligning node*: aligns data streams by timestamps using linear interpolation and saves synchronized data.

Next, we establish the mapping between the inductive sensor signals and the joint states. After alignment, we obtain paired samples $\{(\mathbf{z}_t, \mathbf{q}_t)\}_{t=1}^N$, where $\mathbf{z}_t$ denotes the inductive sensor output and $\mathbf{q}_t$ denotes the corresponding joint angles measured by the vision pipeline.

TABLE II
RANGE OF MOTION (ROM) OF SPIRAL HAND UNDER ACTUATION.

Joint type	Motion	$q_{\min}$ (deg)	$q_{\max}$ (deg)
DIP	Flex./Ext.	-10	90
PIP	Flex./Ext.	-15	100
MCP	Flex./Ext.	-20	90
MCP	Abd./Add.	-10	10
CMC	Flex./Ext.	0	80
CMC	Abd./Add.	0	90
CMC	Axial rot.	0	90

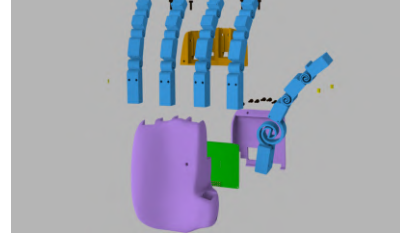


Fig. 6. Exploded view of the hand assembly.

To perform higher-quality mapping, a lightweight multi-layer perceptron (MLP) that directly predicts joint states from sensor readings:

$$\hat{\mathbf{q}} = f_{\text{MLP}}(\mathbf{z}) \quad (7)$$

is preferred. The network is trained to minimize the mean squared error between $\hat{\mathbf{q}}$ and $\mathbf{q}$ in the aligned calibration data in order to better capture coupled nonlinearities and improves robustness when the mapping deviates from a simple monotonic relationship.

E. Hand Platform Implementation

To validate the viability of the PDS joint, an open-source 11-DoF dexterous hand platform is developed. The design details including joint configuration, fingers, palm, and actuation distribution are illustrated below.

a) Joint configuration: The joint geometries are defined through the parameter set $\mathbf{s}$. With these parameters, the resulting range of motion (RoM) for each joint is summarized in Table II and is comparable to that of the human hand.

Notably, the DIP and PIP joints in each finger are driven by a single tendon in an underactuated manner. Although the DIP and PIP joints share a similar RoM in our design, their motion is *sequentially coupled*. During flexion, the PIP joint bends first, followed by the DIP joint, which improves stability during power grasps while maintaining fingertip adaptability.

b) Modular finger design: The four non-thumb fingers share an identical module to simplify fabrication and maintenance. Each finger module integrates the PDS joints with the corresponding inductive sensing components, enabling a standardized bill of materials (BoM) and consistent calibration procedures across fingers. This design reduces the diversity of parts and allows damaged modules to be replaced in minutes. An exploded view of the hand assembly is shown in Fig. 6.

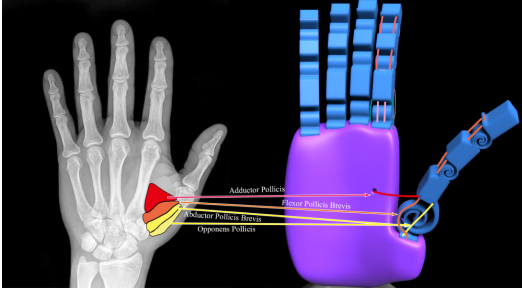


Fig. 7. Specialized actuation distribution of the thumb.

c) Palm design: In addition to serving as a mounting base, the geometry of the palm largely determines the grasp envelope, thumb opposition workspace, and overall manipulation capability. Inspired by human hand anatomy, the palm possesses a human-like curvature profile to support stable power grasps and to increase the effective contact area in grasps. The palm structure is designed to accommodate tendon routing and the sensor wiring while keeping the assembly compact and convenient for maintenance.

d) Efficient actuation distribution: Rather than fully actuating every joint, we allocate independent actuation to critical DoFs and underactuate others to reduce cost. Specifically, the MCP flexion/extension of index and middle fingers is independently actuated to provide precise control of finger posture during pinch and tool use. The DIP and PIP joints are designed to flex in a coupled manner under a single actuation input, reducing actuation complexity while preserving an effective flexion motion. This coupling also improves robustness in contact by allowing the distal phalanges to self-adjust as they wrap around objects.

The ring and pinky fingers are designed with complete underactuated flexion. This design reduces actuator count while enabling robust power grasps and enveloping grasps, where the ring and pinky fingers primarily form grasp closure rather than precise fingertip motions.

The actuation of the thumb is specialized to enable complex opposition behaviors (Fig. 7). We under-actuate the IP, MCP, and CMC joint in flexion, yielding a coordinated motion commonly used in grasp closure. In addition, an independent tendon realizes the function of the adductor pollicis, and another independent tendon combines the function of the opponens pollicis and abductor pollicis brevis. This actuation distribution preserves not only the ability of the thumb to perform power grasps, but also its ability to perform complex manipulation tasks.

IV. EXPERIMENTS

We evaluate the viability of the PDS joint by addressing the following questions:

(Q1) Can the PDS parameters systematically shape the directional stiffness across DoFs? (A)

(Q2) Can embedded inductive sensing provide reliable signals under large deformation? (B)

(Q3) Can learning-based calibration map raw signals to joint states accurately? (C)

(Q4) Do these capabilities enable safe and robust contact-rich demonstrations on a hand platform? (D)

A. Common Setup and Definition

a) Samples and fabrication: All of the samples are 3D printed via Fused Deposition Modeling (FDM) printers under identical settings and share the same Thermoplastic Urethane (TPU 68D) filament. The parameter sets adopted in the samples are summarized in Table III.

b) DoF and coordinate definitions: We refer to the DoFs of the joints as: flexion/extension (FE), abduction/adduction (AA), and pronation/supination (PS). The flexion, abduction, and pronation is defined as the positive direction of motion. All of the zero-positions refer to the initial state of the printed joints unless otherwise noted.

c) Angle and torque measurement: Joint angles were measured using ArUco-based pose estimation at 30 Hz. For stiffness characterization, we applied joint torques using a constant moment-arm mechanism. The resulting forces were measured with a high-resolution dynamometer, which was motor-driven to maintain a constant force unless otherwise noted.

B. Mechanical Characterization

1) Flexion/Extension Stiffness Characterization:

a) Protocol: The torque-angle relationship of the PDS joints during flexion was measured with the setup shown in Fig. 8. Notably, measuring the flexion characterization of a PDS joint with the asymmetric ratio λ is comparable to the extension characterization of the joint with $\lambda' = 1/\lambda$.

b) Results: As shown in Fig. 9, increasing $\theta_{\max}$ and t_0 shifts the torque profiles upward, while increasing slope k tends to reduce stiffness due to larger gaps between the coils. In particular, the asymmetry ratio λ shows a non-monotonic relationship with the actuating torque: when $\lambda > 1$, the

TABLE III
THE PARAMETER SETS FOR THE EXPERIMENTS.

ID	$\theta_{\max}$ (rad)	t_0 (mm)	k_1	k_2	$\lambda=k_2/k_1$
A1	1.5π	1.4	0.8	0.8	1.000
A2	1.75π	1.4	0.8	0.8	1.000
A3	2.0π	1.4	0.8	0.8	1.000
A4	2.25π	1.4	0.8	0.8	1.000
A5	2.5π	1.4	0.8	0.8	1.000
K1	2.5π	1.4	0.6	0.6	1.000
K2	2.5π	1.4	0.7	0.7	1.000
K3	2.5π	1.4	0.8	0.8	1.000
K4	2.5π	1.4	0.9	0.9	1.000
K5	2.5π	1.4	1.0	1.0	1.000
T1	2.5π	0.8	0.8	0.8	1.000
T2	2.5π	1.0	0.8	0.8	1.000
T3	2.5π	1.2	0.8	0.8	1.000
T4	2.5π	1.4	0.8	0.8	1.000
T5	2.5π	1.6	0.8	0.8	1.000
R1	2.5π	1.4	0.8	0.70	0.875
R2	2.5π	1.4	0.8	0.75	0.938
R3	2.5π	1.4	0.8	0.80	1.000
R4	2.5π	1.4	0.8	0.85	1.062
R5	2.5π	1.4	0.8	0.90	1.125

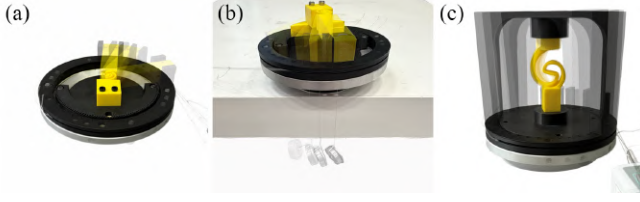


Fig. 8. (a) Experimental setup for mechanical characterization in flexion/extension. (b) Experimental setup for mechanical characterization in abduction/adduction. (c) Experimental setup for mechanical characterization in pronation/supination.

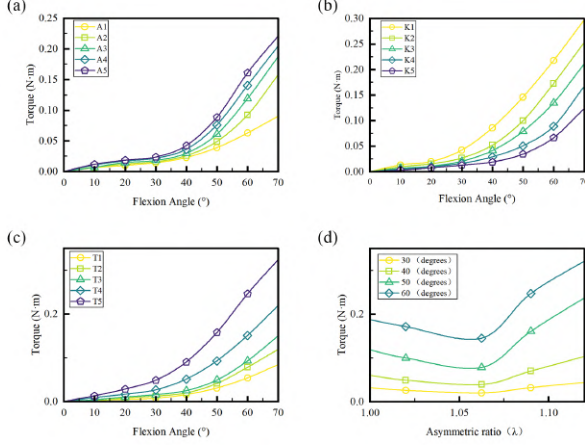


Fig. 9. (a) Measured flexion/extension torque-angle curves with varying $\theta_{\max}$. (b) Measured flexion/extension torque-angle curves with varying k . (c) Measured flexion/extension torque-angle curves with varying t_0 . (d) Processed flexion/extension torque-angle curves with varying λ .

torque may decrease slightly before rising rapidly, with a local minimum observed near $\lambda \approx 1.06$ in our measurements.

2) Abduction/Adduction Stiffness Characterization:

a) Protocol: The lateral-stiffness-angle relationship of the PDS joint was measured under constant lateral force (0.5N) from a hanging weight at varying flexion angles, using the setup shown in Fig. 8.

b) Results: Fig. 10 shows that in all samples, the lateral stiffness increases with the flexion angle. The profiles shift downward as $\theta_{\max}$ and k increase, while increasing t_0 shifts the profiles upward. When $\theta_{\max}$ increases to 2.5π , the lateral stiffness increases dramatically with increasing flexion, which is beneficial to provide compliance during the approach while maintaining rigidity in grasp. Finally, as the asymmetry ratio λ increases over 1.00, the lateral stiffness first increases, reaching a maximum at around $\lambda = 1.25$, and then presents another non-monotonic behavior, with a minimum at approximately $\lambda = 1.55$. This nontrivial relationship indicates that the asymmetric design effectively improves the lateral stiffness in a specific range, and highlights the importance of fine parameter tuning rather than simply maximizing λ .

3) Pronation/Supination Stiffness Characterization:

a) Protocol: Measurements of torsional torque versus twist angle for the thumb CMC joint were conducted with

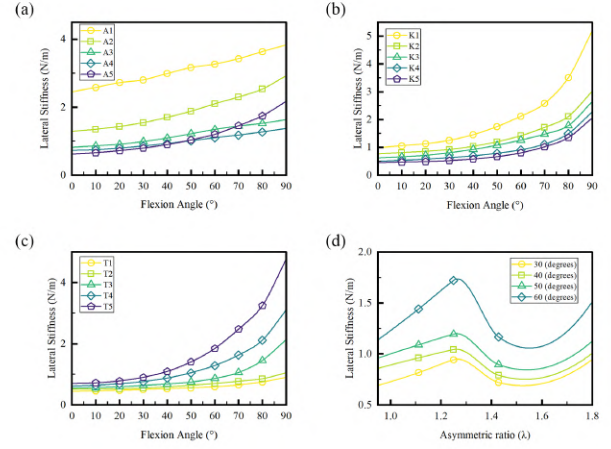


Fig. 10. (a) Measured lateral stiffness versus deflect angle with varying $\theta_{\max}$. (b) Measured lateral stiffness versus deflect angle with varying k . (c) Measured lateral stiffness versus deflect angle with varying t_0 . (d) Measured lateral stiffness versus deflect angle with varying λ .

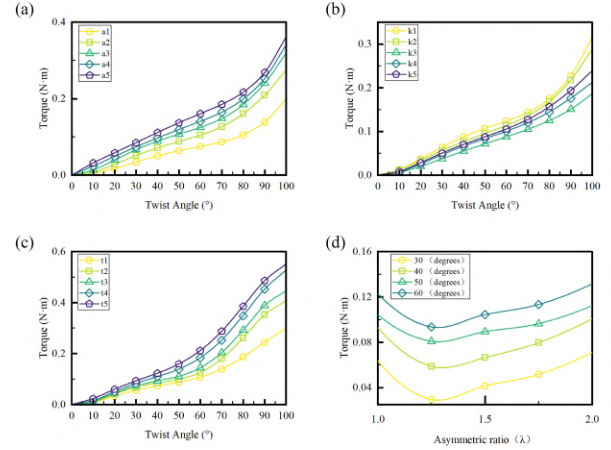


Fig. 11. (a) Measured pronation/supination torque-angle curves with varying $\theta_{\max}$. (b) Measured pronation/supination torque-angle curves with varying k . (c) Measured pronation/supination torque-angle curves with varying t_0 . (d) Measured pronation/supination torque-angle curves with varying λ .

the setup illustrated in Fig. 8.

b) Results: Measured torque correlates positively with twist angle, as shown in Fig. 11. Increasing $\theta_{\max}$ and t_0 shifts the profiles upward, while k exhibits a non-monotonic effect. A similar non-monotonic trend of torque profiles is observed with respect to λ to that in the flexion/extension experiment. The local minimum appears around $\lambda \approx 1.26$ in this case.

Based on these findings, we can decide the parameters for the final hand design. For DIP and PIP joints, we pursue high lateral stiffness, while for the MCP joint, reduced lateral stiffness is preferred to enable easier adduction/abduction. For the CMC joint, we prioritize easier axial rotation and lower lateral stiffness for opposition.

4) Repeatability Under Cyclic Loading:

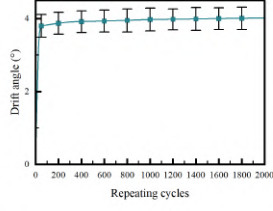


Fig. 12. Drift angle versus cycle count in the cyclic loading test.

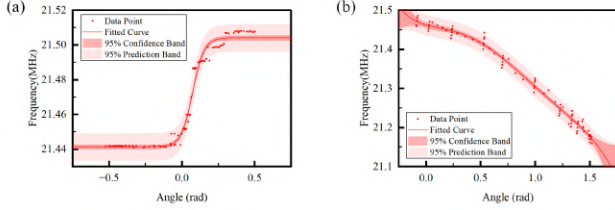


Fig. 13. fitted curves of raw inductive signals versus joint angle for PIP flexion and MCP abduction.

a) Protocol: To evaluate long-term repeatability under repeated large deformation, we conducted a cyclic loading test on the PDS joint. Each cycle consisted of flexing the joint from its natural position to a prescribed maximum angle $q_{\max}$, and releasing the load to observe the drift angle from zero-position. The joint was actuated at a frequency of 2 Hz, and the test was repeated for 2000 cycles under identical conditions at room temperature.

b) Results: As shown in Fig. 12, the drift angle quickly increases to around 3.8° , and then remained at a relatively stable offset from zero-position. Based on this observation, we deliberately enlarged the extension angle at zero-position slightly to enable the desired ROM.

C. Proprioception and Calibration Evaluation

1) Raw Inductive Signal Characterization:

a) Protocol: To illustrate the effectiveness of the inductive sensing system, the raw-signal-bending-angle relationships of the PIP and MCP joints are presented in Fig. 13.

b) Results: The single-DoF PIP joint demonstrated an approximately linear relationship between the signal and the joint angle. However, in AA motion of MCP joints, where the distance d between the inductor and the copper target is significantly narrowed, signal saturation is observed at large angles.

2) Joint-State Mapping: Fitting vs. MLP:

a) Protocol: To evaluate the effectiveness of the MLP calibration pipeline, we present the parity plots of ground truth angles and estimated angles in the most difficult AA motion. Then, we compare the learning-based method with conventional curve fitting method through the plots.

b) Results: As Fig. 14 suggests, the MLP method possesses a MAE of 0.051 and $RMSE$ of 0.067, outperforming the conventional curve fitting method ($MAE=0.125$, $RMSE=0.172$).

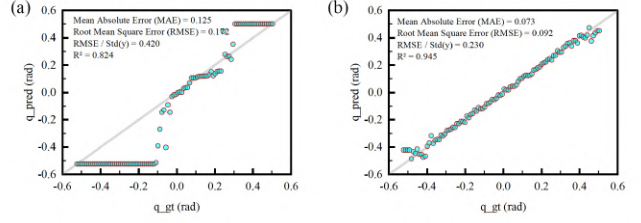


Fig. 14. Joint-state mapping results in the challenging MCP abduction/adduction case: (a) conventional fitting versus (b) MLP-based method.

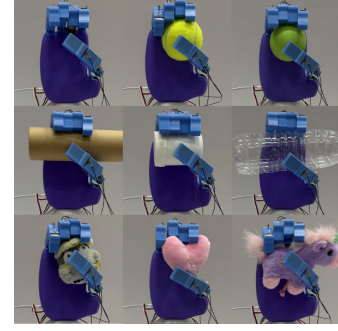


Fig. 15. Adaptive grasping demonstrations with various kinds of objects

D. Demonstration on Hand Platform

1) Adaptive grasping:

a) Protocol: We evaluate grasping robustness using the hand platform equipped with PDS joints. A set of N_o everyday objects with diverse geometries and sizes was used, including spherical, cylindrical, and irregular stuffed toys. For each trial, the hand started from a fixed initial pose, closed the fingers under the same position sequence command. The demonstrations relied on the inductive proprioception system to observe joint states during grasps. Each object was tested for 10 trials, and a stable grasp maintained over 30 seconds is considered successful.

b) Results: The compliant hand integrated with the PDS joint and the inductive sensing system achieved full success in the grasping test, and the synchronized visualization realized in PyBullet is presented along with the grasping poses in Fig. 15.

2) Contact-rich Demonstrations with Safety:

a) Protocol: To further investigate the effectiveness of the design, we implemented the Kapandji test, and two contact-rich manipulation tasks involving pinch grasp and thumb opposition. Firstly, we placed a syringe between the index and the middle fingers in motion, and let the hand pinch grasp it and push the plunger upward. Next, we let the hand take hold of an electric screwdriver in motion and press the button to drive it. In addition, the safety of compliant joints is intuitively illustrated in these tasks.

b) Results: Our hand platform scored 10/10 in the Kapandji test, and successfully performed these tasks as shown in Fig. 16, demonstrating the dexterity provided by the PDS joint. Fig. 17 illustrated how the PDS joint absorbs

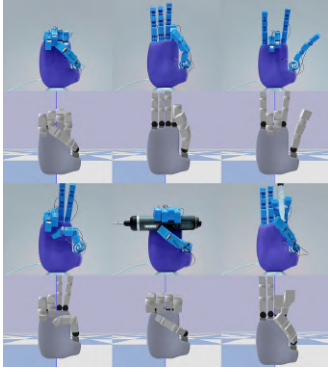


Fig. 16. Contact-rich manipulation demonstrations and synchronized PyBullet visualization.



Fig. 17. Illustration of impact absorption and safe human-involved interaction enabled by the compliant PDS joint.

impact and collision in use, ensuring safety in human-involved tasks.

V. CONCLUSION

This paper presented the *PDS joint*, a parametric double-spiral compliant joint designed for joint-specific-stiffness designs in dexterous hands. By parameterizing double-spiral geometries, the proposed design provides a unified geometric paradigm to tailor stiffness distributions across flexion/extension, abduction/adduction, and pronation/supination, enabling both large-stroke motion and improved off-axis support within selected parameter ranges. To address the state-estimation challenge inherent in compliant mechanisms, we integrated compact inductive proprioception into the joint and introduced a practical learning-based calibration pipeline using ArUco-supervised data, which improved joint-state mapping accuracy over conventional curve fitting.

Comprehensive experiments validated the approach from component to system levels. We investigated the stiffness characteristics over key geometric parameters and demonstrated repeatability under cyclic loading. The inductive sensing signals exhibited repeatable trends with joint configuration, and the learning-based calibration achieved lower estimation error in challenging motions. Finally, integration on an open-source dexterous hand demonstrated safe and robust contact-rich tasks, including adaptive grasping and human-involved tool operations.

Future work will extend long-term fatigue evaluation under broader loading conditions, improve sensor placement to mitigate saturation at extreme poses, and leverage the calibrated proprioception for closed-loop manipulation and

higher-level-policy deployment such as Imitation Learning (IL) and Reinforcement Learning (RL).

REFERENCES

- [1] R. R. Ma, L. U. Odhner, and A. M. Dollar, "A modular, open-source 3d printed underactuated hand," in *2013 IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 2722–2727.
- [2] D. Rus and M. T. Tolley, "Design, fabrication and control of soft robots," *Nature*, vol. 521, no. 7553, pp. 467–475, 2015.
- [3] M. Naves, D. M. Brouwer, and R. G. K. M. Aarts, "Building block-based spatial topology synthesis method for large-stroke flexure hinges," *Journal of Mechanisms and Robotics*, vol. 9, no. 4, p. 041006, 2017.
- [4] S. Kim, E. Sung, and J. Park, "ARC joint: Anthropomorphic rolling contact joint with kinematically variable torsional stiffness," *IEEE Robotics and Automation Letters*, vol. 8, no. 3, pp. 1810–1817, 2023.
- [5] R. Cruz and A. J. Hetherington, "Identification of a tetrahedral apical cell preserved within a fossilized fern fiddlehead," *Current Biology*, vol. 35, no. 2, pp. 383–390.e2, 2025.
- [6] P. J. Grubb and R. V. Jackson, "The adaptive value of young leaves being tightly folded or rolled on monocotyledons in tropical lowland rain forest: an hypothesis in two parts," *Plant Ecology*, vol. 192, pp. 317–327, 2007.
- [7] M. Jafarpour, S. Gorb, and H. Rajabi, "Double-spiral: a bioinspired pre-programmable compliant joint with multiple degrees of freedom," *Journal of the Royal Society Interface*, vol. 20, p. 20220757, 2023.
- [8] M. Jafarpour, M. Aryayi, S. N. Gorb, and H. Rajabi, "Double-spiral as a bio-inspired functional element in engineering design," *Scientific Reports*, vol. 14, p. 29225, 2024.
- [9] K. Junge and J. Hughes, "Spatially distributed biomimetic compliance enables robust anthropomorphic robotic manipulation," *Communications Engineering*, vol. 4, no. 1, p. 76, 2025.
- [10] K. Suzumori and H. Nabae, "Review of flexible/stretchable sensors for soft robot," *Journal of Robotics and Mechatronics*, vol. 37, no. 1, pp. 8–12, 2025.
- [11] B. Brajon, L. Lugani, and G. Close, "Hybrid magnetic-inductive angular sensor with 360^{circ} range and stray-field immunity," *Sensors*, vol. 22, no. 6, p. 2153, 2022.
- [12] Infineon Technologies, "Inductive sensing design guide (an219207)," Infineon Technologies, Tech. Rep., 2024, application note.
- [13] Texas Instruments, "High-resolution low latency and low-power absolute inductive angle encoder reference design (tida-010961)," Texas Instruments, Tech. Rep., 2025, design Guide, released Sep 23, 2025. [Online]. Available: <https://www.ti.com/tool/TIDA-010961>
- [14] L. A. Garcia Rodriguez, M. Naves, and D. M. Brouwer, "3d-printed flexure-based finger joints for anthropomorphic hands," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 1437–1442.
- [15] Y. Kim, J. Lee, and J. Park, "Compliant joint actuator with dual spiral springs," *IEEE/ASME Transactions on Mechatronics*, vol. 18, no. 6, pp. 1839–1844, 2013.
- [16] G. Radaelli and J. L. Herder, "Study on the large-displacement behaviour of a spiral spring with variations of cross-section, orthotropy and prestress," *Mechanical Sciences*, vol. 9, pp. 337–348, 2018.
- [17] J. Korenblik, "Minimal sensing approach of an underactuated flexure based gripper for agri-food applications," M.Sc. Thesis, University of Twente, Aug. 2021.
- [18] B. Aksoy, Y. Hao, G. Grasso, K. M. Digumarti, V. Cacucciolo, and H. Shea, "Shielded soft force sensors," *Nature Communications*, vol. 13, no. 1, p. 4649, 2022.
- [19] Renesas Electronics, "Inductive position sensors," Product page, 2026, accessed 2026-03-06. [Online]. Available: <https://www.renesas.com/en/products/sensor-products/inductive-position-sensors>
- [20] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez, "Automatic generation and detection of highly reliable fiducial markers under occlusion," *Pattern Recognition*, vol. 47, no. 6, pp. 2280–2292, 2014.
- [21] Texas Instruments, "LDC1612, LDC1614: Multi-channel 28-bit inductance to digital converter (ldc) for inductive sensing, datasheet (rev. a)," <https://www.ti.com/product/LDC1614>, 2018, accessed 2026-03-03.
- [22] Y. Peng, H. Wu, Z. Wang, Y. Wang, and H. Wang, "A soft inductive bi-modal sensor for proprioception and tactile sensing of soft machines," *Soft Robotics*, vol. 11, no. 6, pp. 1055–1067, 2024.